

Graph Search Algorithms

Kanishk Goel

14 November 2025

1 Free Primitives

Treat any algorithm with linear running time as something we can essentially use for free.

2 Generic Graph Search

Input: An undirected/directed graph $G = (V, E)$, and a starting vertex $s \in V$

Output: Identify the vertices of V reachable from s in G .

Algorithm:

Input: graph $G = (V, E)$ and vertex $s \in V$

Postcondition: a vertex is reachable from s if and only if it is marked as "explored"

mark s as explored and all others unexplored

while $\exists (v, w) \in E$ with v explored and w unexplored **do**

 choose some edge (v, w)

 mark w as explored

end while

3 Breadth First Search

Explores layer by layer, Layer 0 being s itself.

mark s as explored, all other vertices are unexplored

Q := a queue data structure, init with s

while Q is not empty **do**

 remove the vertex from the front of Q , call it v

for each edge (v, w) in v 's adjacency list **do**

if w is unexplored **then**

 mark w as explored

 add w to the end of queue

4 Depth First Search

Takes one path and only backtracks when absolutely necessary.